Synchronous Generators/ Alternators:

The machines generating a.c e.m.f are called alternators or synchronous generators

Principle of operation:

Commonly used machine for generation of electrical power is known as synchronous generator. It is also known as alternator since it generates alternating voltage. Alternators are use in generating stations to generate electric power.

A synchronous generator works on the principle of Faraday's law of electromagnetic induction i.e. when a conductor is moved by a prime mover in a magnetic field in a direction so as to cut across the magnetic lines of flux; a voltage is induced in it. This induced voltage is said to be dynamically induced since it is caused by the relative motion of the conductor and magnetic field. The magnitude of the induced E.M.F in the conductor is given by

$$E = Blv\sin\theta$$

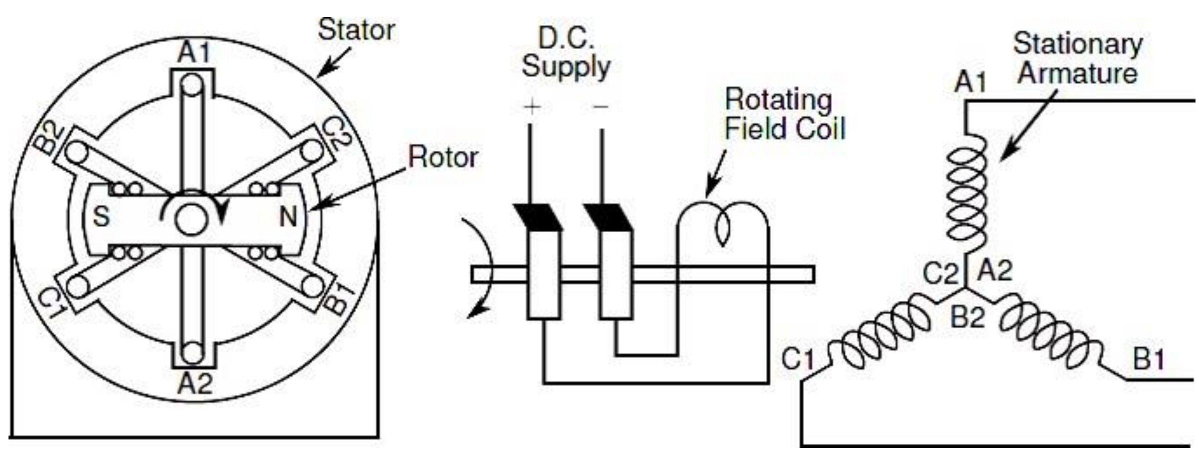
Where, l=length of the portion of conductor within the magnetic field

B= Magnetic flux density

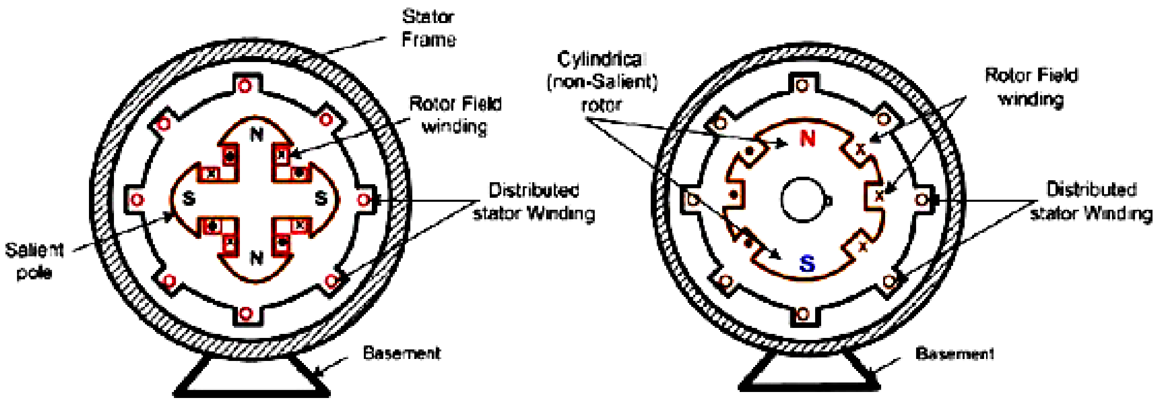
v=velocity of the conductor

θ= angle between the direction of movement of conductor and direction of magnetic flux

The direction of the induced emf is given by Fleming's right hand rule.In all commercial synchronous generators, a three phase winging is provided in the stator slots and a field coil is placed on the rotor as shown in figure below. A D.C supply is given to the field coil through the brushes and slip rings. The rotor is rotated by a prime mover to cause the relative motion of conductor with respect to the magnetic flux lines.



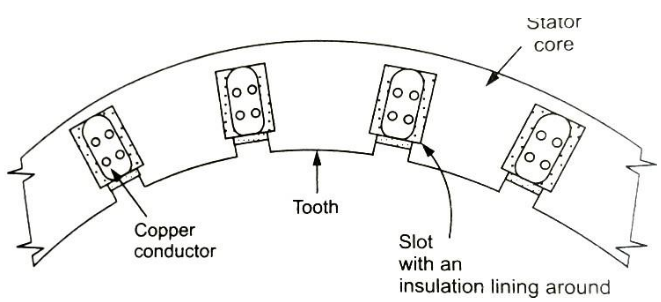
Types and Constructional Features:
A synchronous generator has two main parts, viz. stator and rotor



Salient-Pole

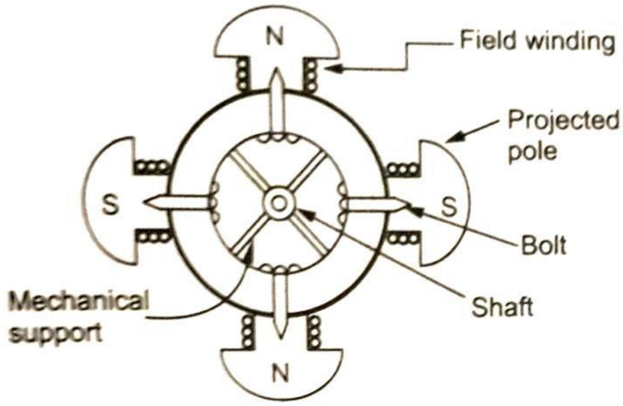
Non-Salient-Pole

Stator: Stator is a stationary part. This consists of a core and slots to hold the armature winding in which A.C voltage is induced. The stator is built using a cast steel frame, within this frame, laminations of silicon steel having slots in the inner periphery are stacked to form the armature core. Each lamination is insulated from each other with varnish or paper. The laminated construction is basically used to reduce the eddy current losses. Within the slots of armature core, conductors are placed and interconnected to form the armature winding. The three phase armature winding is evenly distributed. Coils of armature winding are made from insulated copper conductors.

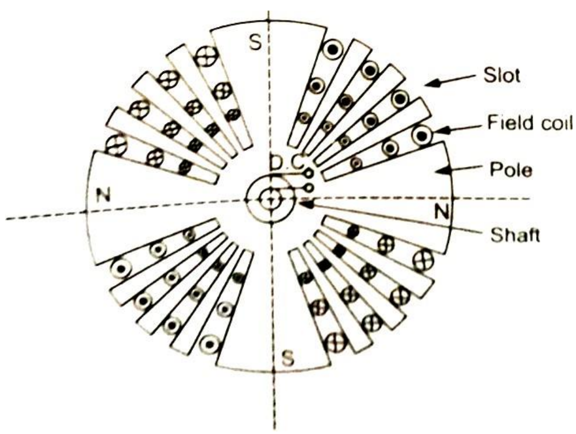


Rotor:
There are two types of rotors used in alternator
1. Salient pole type
2. Non-salient pole type or smooth cylindrical type

Salient pole type rotor:
This is also called projected pole type as all the poles are projected out from the surface of the rotor. The poles are built of thick steel laminations. The poles are bolted to the rotor as shown in the figure. The field winding is provided on the pole shoe. These rotos have large diameters and small axial lengths. The limiting factor for the size f the rotor is the centrifugal force acting on the rotating member of the machine. As the mechanical strength of the salient pole type rotor is less, this is preferred foe low speed alternators ranging from 125 r.p.m to 500 r.p.m.



Non-salient pole type or smooth cylindrical type rotor:
This is also called projected pole type rotor. The rotor consists of smooth solid steel cylinder, having number of slots to accommodate the field coils as shown in the figure. The slots are covered at the top with the help of steel or manganese wedges. The unslotted portions of the cylinder itself act as the poles. The poles are not projecting out and the surface of the rotor is smooth which maintains uniform air gap between stator and rotor. These rotors have small diameters and large axial lengths. This keeps the peripheral speed within limits. The main advantage of this typs is that these are mevhanically very strong and thus preferred for high speed alternators rating between 1500 to 3000 r.p.m



Frequency of induced E.M.F:
When a conductor moves past a pair of poles, one cycle of sinusoidal voltage is completed.
If P=total no. of poles in the machine, then,
Number of cycles per revolution =P/2
If N = R.P.M of the motor , then rotor makes N/60 R.P.S
Hence the frequency of the induced E.M.F is
$f=(P/2) \times (N/60) = PN/120$ Hz

Asynchronous machines

Asynchronous motor is a machine whose rotor rotates at the speed less than the synchronous speed. Asynchronous machines are classified into two types based on the supply.

1. Three phase induction motors
2. Single phase induction motors